

Course Contents

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Course Contents

Unit-01:Introduction to Computer

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Course Contents

Unit-01:Introduction to Computer (4 Hrs.)

- **Introduction of Computer**

Questions:

- What is Computer? Draw the block diagram of Digital computer and explain it's components in brief.

Introduction to computer

Computer:

- The term computer is derived from the word compute. The word compute means to calculate.
- A computer is an electronic machine that accepts data from the user, processes the data by performing calculations and operations on it, and generates the desired output results.
- Computer performs both simple and complex operations, with speed and accuracy.

Introduction to computer

Computer:

- A computer is an electronic device, operating under the control of instructions stored in its own memory that can accept data (input), process the data according to specified rules, produce information (output), and store the information for future use.
- A computer is a programmable machine. The two principal characteristics of computer are:
 - It responds to a specific set of instructions in a well-defined manner.
 - It can execute a prerecorded list of instructions (program)

Introduction to computer

Computer:

- All general-purpose computers require the following hardware components.
- Central Processing Unit (CPU) : The heart of the computer, the component that actually executes instructions.
- Memory: Enables a computer to store, at least temporarily, data and programs.
- Input Device: Input devices are the devices which are used to feed programs and data to the computer. The input system connects the external environment with the computer system.

Introduction to computer

Computer:

- **Output Device:** The output devices give the results of the process and computations to the outside world. The output units accept the results produced by the computer, convert them into a human readable form and supply them to the users. The more common output devices are printers, plotters, display screens, magnetic tape drives etc.
- **Mass storage device:** Allows a computer to permanently retain large amounts of data. Common mass storage devices include disk drives and tape drives.

Introduction to computer

Block Diagram of Computer:

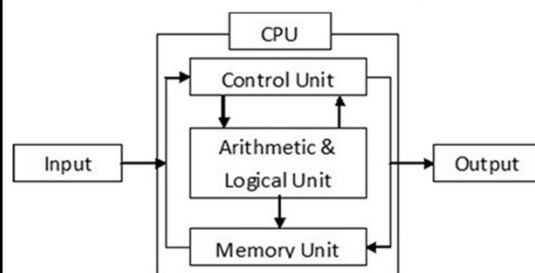
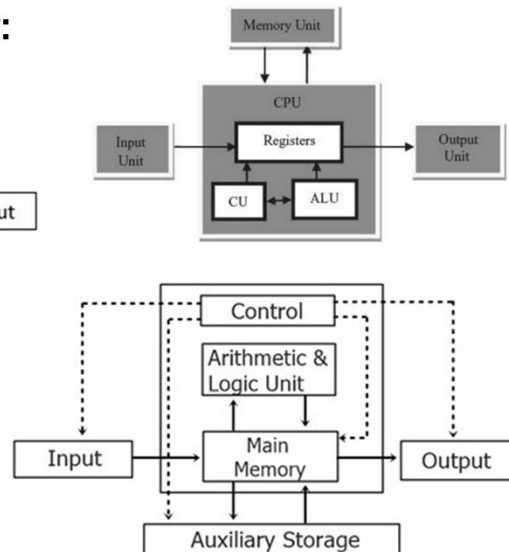


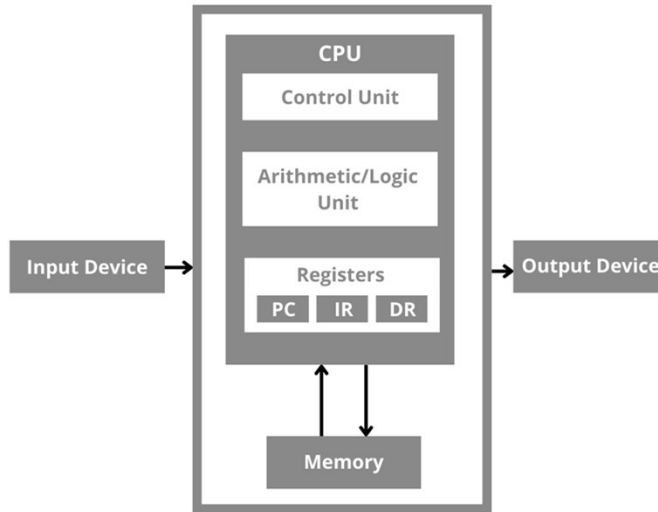
Fig. Block Diagram of Computer



Block Diagram of Computer

Introduction to computer

Block Diagram of Computer:



Introduction to computer

8085 Block Diagram:

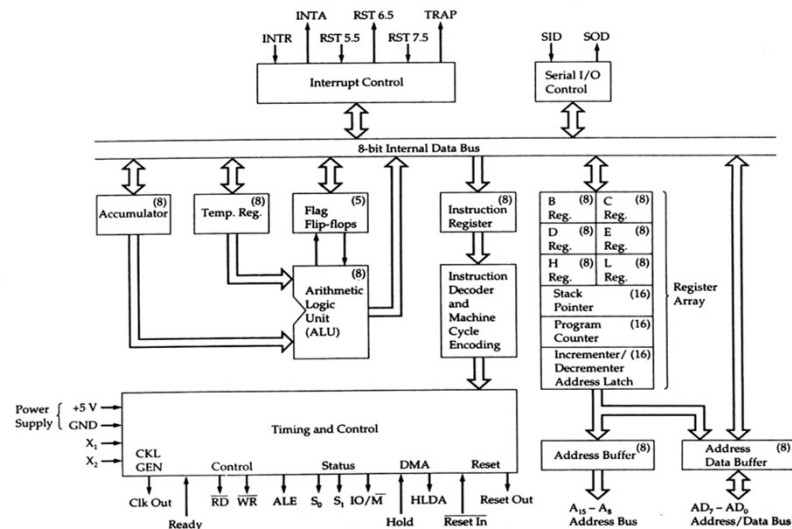


Figure: Intel 8085 CPU Block Diagram

Characteristics of Computer

Characteristics of Computer:

1. Word length
2. Speed
3. Storage
4. Accuracy
5. Versatility
6. Automation
7. Deligence

Characteristics of Computer

Characteristics of Computer:

1. Word length:

Difference between bit, nibble, byte and wordlength.

Word length is the computing power of computer and varies as 8, 16, 32 or 64 bits. Longer the word length, more powerful the computer is.

2. Speed: time taken to perform any task by computer is called the speed of the computer. The speed of computer is measured in terms of micro or nano-second.

Milliseconds(ms) = $1/1,000$; microseconds(μ s)= $(1/1,000,000)$, Nanoseconds(ns)= $(1/1,000,000,000)$,
Picoseconds(ps) = $(1,000,000,000,000)$

Characteristics of Computer

Characteristics of Computer:

3. Storage:

- Digital data storage is essentially the recording of digital information in a storage medium, typically by electronic means.
- Computers have their own main memory and auxiliary memory storage systems which computer use when needed.
- The storage capacity is measured in terms of

Symbol	Prefix
k	kilo
M	mega
G	giga
T	tera

Characteristics of Computer

Characteristics of Computer:

3. Storage:

Symbol	Prefix	SI Meaning	Binary meaning	Size difference
k	kilo	$10^3 = 1000^1$	$2^{10} = 1024^1$	2.40%
M	mega	$10^6 = 1000^2$	$2^{20} = 1024^2$	4.86%
G	giga	$10^9 = 1000^3$	$2^{30} = 1024^3$	7.37%
T	tera	$10^{12} = 1000^4$	$2^{40} = 1024^4$	9.95%
P	peta	$10^{15} = 1000^5$	$2^{50} = 1024^5$	12.59%
E	exa	$10^{18} = 1000^6$	$2^{60} = 1024^6$	15.29%
Z	zetta	$10^{21} = 1000^7$	$2^{70} = 1024^7$	18.06%
Y	yotta	$10^{24} = 1000^8$	$2^{80} = 1024^8$	20.89%

Characteristics of Computer

Characteristics of Computer:

4. Accuracy:

- Computer is the accurate machine with high accuracy and every calculation is performed with the same accuracy.
- It can perform large number of task without errors but if we feed wrong data, it returns the same wrong information called GIGO(Garbage In Garbage Out).

5. Versatility:

Versatile means, computer can not only perform calculation but same computer can perform many different types of Job depend upon the different program fed to it.

Characteristics of Computer

Characteristics of Computer:

6. Automation:

- Automation is the technology by which a process or procedure is performed with minimal human assistance.
- Computers are capable for automation, provided they are programmed correctly.
- Computers can proceed on its own till its completion.

Characteristics of Computer

7. Diligence:

- Diligence of computer is the capacity of performing same task repeatedly multiple times without feeling tiredness, boring, lack of concentration and fatigue with same speed and accuracy.

Characteristics of Computer

7. Diligence:

- Human being suffer from weakness like tiredness, lack of concentration etc. Human have feelings they become sad, depressed, bored, and negligent and will reflect on the work they do.
- Human beings can't perform the same or similar tasks over and over again with the same precision, accuracy and enthusiasm as the first time. This will effect the performance.

Being a machine, a computer doesn't have any of these human weaknesses.

Introduction to computer

History Vs Generation of Computer ?

Introduction to computer

History of Computer:

History of computing is a timeline of how computers evolved from their humble beginnings to the machines of today that surf the Internet, play games and stream multimedia in addition to crunching numbers.

<https://www.livescience.com/20718-computer-history.html>

https://wikieducator.org/History_of_Computer_Development%26_Generation_of_Computer

Introduction to computer

Generations of Computer:

- Computer generation is a change in technology a computer is/was being used.
- Initially, the generation term was used to distinguish between varying hardware technologies.
- But nowadays, generation includes both hardware and software, which together make up an entire computer system.

https://www.webopedia.com/DidYouKnow/Hardware_Software/FiveGenerations.asp

https://wikieducator.org/History_of_Computer_Development_%26_Generation_of_Computer

Introduction to computer

History of Computer: The key developments that took place till the first computer was developed are as follows:

1. Calculating Machines ABACUS was the first mechanical calculating device for counting of large numbers. The word ABACUS means calculating board. It consists of bars in horizontal positions on which sets of beads are inserted. The horizontal bars have 10 beads each, representing units, tens, hundreds, etc.

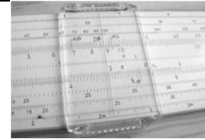


2. Napier's Bones was a mechanical device built for the purpose of multiplication in 1617 by an English mathematician John Napier.

Introduction to computer

History of Computer:

3. Slide Rule was developed by an English mathematician Edmund Gunter in the 16th century. Using the slide rule, one could perform operations like addition, subtraction, multiplication and division. It was used extensively till late 1970s.

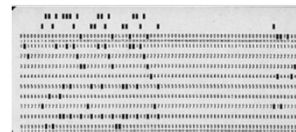


4. Pascal's Adding and Subtraction Machine was developed by Blaise Pascal. It could add and subtract. The machine consisted of wheels, gears and cylinders.
5. Leibniz's Multiplication and Dividing Machine was a mechanical device that could both multiply and divide. The German philosopher and mathematician Gottfried Leibniz built it around 1673.

Introduction to computer

History of Computer:

6. Punch Card System was developed by Jacquard to control the power loom (weave cloth) in 1801. He invented the punched card reader that could recognize the presence of hole in the punched card as binary one and the absence of the hole as binary zero. The 0s and 1s are the basis of the modern digital computer.



Introduction to computer

History of Computer:

7. Babbage's Analytical Engine An English man Charles Babbage built a mechanical machine to do complex mathematical calculations, in the year 1823. The machine was called as difference engine. Later, Charles Babbage and Lady Ada Lovelace developed a general-purpose calculating machine, the analytical engine. Charles Babbage is also called the father of computer.
8. Hollerith's Punched Card Tabulating Machine was invented by Herman Hollerith. The machine could read the information from a punched card and process it electronically.

Introduction to computer

Generations of Computer:

- Computer generation is a change in technology a computer is/was being used.
- Initially, the generation term was used to distinguish between varying hardware technologies.
- But nowadays, generation includes both hardware and software, which together make up an entire computer system.

https://www.webopedia.com/DidYouKnow/Hardware_Software/FiveGenerations.asp

https://wikieducator.org/History_of_Computer_Development_%26_Generation_of_Computer

Introduction to computer

Generations of Computer:

Currently, there are 5 generations of computer and generations of computer are differentiated in terms of

- Technology (hardware and software),
- Computing characteristics (speed),
- Physical appearance(Size)
- Applications.

Introduction to computer

Generations of Computer:

First Generation: Vacuum Tubes (1940-1956):

- Used vacuum tubes for circuitry and magnetic drums for memory, and taking up entire rooms. Very expensive to operate and in addition to using a great deal of electricity, the first computers generated a lot of heat, which was often the cause of malfunctions.
- Relied on machine language to perform operations and solve one problem at a time. Would take operators days or even weeks to set-up a new problem. Input was based on punched cards and paper tape, and output was displayed on printouts.
- UNIVAC and ENIAC computers are examples of first-generation computing devices. The UNIVAC was the first commercial computer delivered to a business client, the U.S. Census Bureau in 1951.

Introduction to computer

Generations of Computer:

Second Generation: Transistors (1956-1963):

- Transistor was used in place of vacuum tube, allowing computers to become smaller, faster, cheaper, more energy-efficient and more reliable. Though the transistor still generated a great deal of heat but was a vast improvement over the vacuum tube. Still relied on punched cards for input and printouts for output.
- Moved from binary to assembly languages. High-level programming languages were also being developed at this time, such as early versions of COBOL and FORTRAN. These were also the first computers that stored their instructions in their memory, which moved from a magnetic drum to magnetic core technology.
- The first computers of this generation were developed for the atomic energy industry.

Introduction to computer

Generations of Computer:

Third Generation: Integrated Circuits (1964-1971):

- Integrated circuit was the hallmark of the third generation of computers. Transistors were miniaturized and placed on silicon chips, called semiconductors, which drastically increased the speed and efficiency of computers.
- Used keyboards and monitors instead of punched cards and printouts and interfaced with an operating system, which allowed the device to run many different applications at one time with a central program that monitored the memory.
- Computers for the first time became accessible to a mass audience because they were smaller and cheaper than their predecessors.

Introduction to computer

Generations of Computer:

Fourth Generation: Microprocessors (1971-Present):

- The microprocessor-thousands of integrated circuits were built onto a single silicon chip. The Intel 4004 chip, developed in 1971, located all the components of the computer—from the central processing unit and memory to input/output controls—on a single chip.
- In 1981 IBM introduced its first computer for the home user, and in 1984 Apple introduced the Macintosh.
- Computers became more powerful, they could be linked together to form networks, which eventually led to the development of the Internet.
- Fourth generation computers also saw the development of GUIs, the mouse and handheld devices.

Introduction to computer

Generations of Computer:

Fifth Generation: Artificial Intelligence (Present and Beyond):

- Fifth generation computing devices, based on artificial intelligence, are still in development, though there are some applications, such as voice recognition, that are being used today. The use of parallel processing and superconductors is helping to make artificial intelligence a reality.
- Quantum computation and molecular and nanotechnology will radically change the face of computers in years to come. The goal of fifth-generation computing is to develop devices that respond to natural language input and are capable of learning and self-organization.

Generations of Computer

Generations of Computer:

Following are the main five generations of computers.

S.N.	Generation & Description
1	First Generation The period of first generation: 1942-1954. Vacuum tube based.
2	Second Generation The period of second generation: 1952-1964. Transistor based.
3	Third Generation The period of third generation: 1964-1972. Integrated Circuit based.
4	Fourth Generation The period of fourth generation: 1972-1990. VLSI microprocessor based.
5	Fifth Generation The period of fifth generation: 1990-onwards. ULSI microprocessor based

Generations of Computer

Generations of Computer:

Following are the main five generations of computers.



Vacuum tube

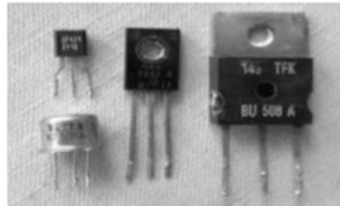


Fig 1.5 Transistors



Fig 1.6 i) Integrated Circuit



ii) Integrated Circuit



Fig. 1.7 VLSI

Questions

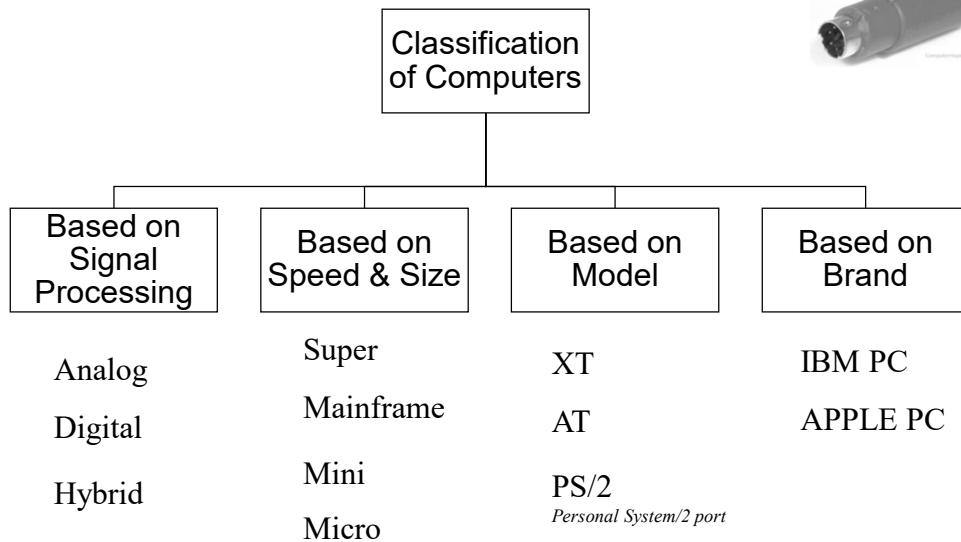
- Explain briefly the developments in computer technology starting from a simple calculating machine to the first computer.
- Describe the first generation computer based on the (a) Hardware (b) Software (c) Computing characteristics (d) Physical appearance, and (e) Their applications.

Introduction to computer

Classification of Computer

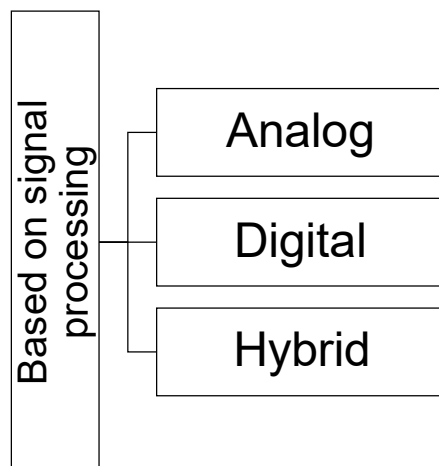
Classification of Computers

Classification of Computers:



Classification of Computers

Classification of Computers based signal processing:



Classification of Computers

Classification of Computers:(Based on Signal Processing)

- **Analog Computer:** Analog computer is that computer, which is use to process continuously varying data. Everything we see and hear is change continuously. This changeable continuous stream of data is called analog data. Analog computer can be used in scientific and industrial applications such as measure the electrical current, frequency and resistance of capacitor, etc.



Classification of Computers

Classification of Computers:

- **Digital Computer:** These are high speed electronic devices. These devices are programmable. They process data by way of mathematical calculations, comparison, sorting etc. They accept input and produce output as discrete signals representing high (on) or low (off) voltage state of electricity. Numbers, alphabets, symbols are all represented as a series of 1s and 0s.

Classification of Computers

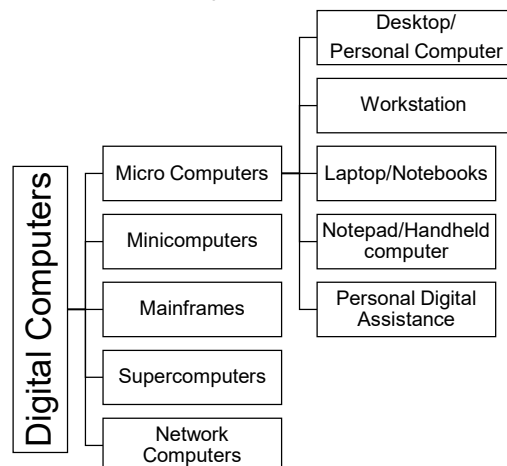
Classification of Computers:

- Hybrid Computer: Hybrid computer is a digital computer that accepts analog signals, converts them to digital and processes them in digital form.
- Hybrid computer processes both analog and digital data.
- Examples: Computer used in hospitals to measure the heartbeat of the patient. Devices used in petrol pump.
- In scientific applications or in controlling industrial processes.
- The first desktop hybrid computing system was the Hycomp 250, released by Packard Bell in 1961.

Classification of Computers

Classification of Computers:

Classification of digital computers on the basis of their capacity to access memory and size are like:



Introduction to computer

Classification of Computers:

Digital Computer: Classification of digital computers on the basis of their capacity to access memory and size are like:

- Micro Computer
 - Desktop / Mini PC / Workstation
 - Laptops/Notebooks
 - Notepad/handheld computer
 - Personal Digital Assistants (PDAs)
- Minicomputers
- Mainframes
- Supercomputers
- Network Computers

Classification of Computers

Classification of Computers based on size:

Super Computer:

- A supercomputer is a type of computer that has the architecture, resources and components to achieve massive computing power. Today's supercomputers consists of tens of thousands of processors that are able to perform billions and trillions of calculations or computations per second.
- As of 2013, IBM Sequoia is the fastest supercomputer to date. It has more than 98,000 processors that allow it to process at a speed of 16,000 trillion calculations per second.

Classification of Computers

Classification of Computers based on size:

Mainframe:

- Mainframe Computers are less costly, small in size and slower in speed than the super computers.
- used as a storage for large database and serve as a maximum number of users simultaneously.
- millions of instructions are executed simultaneously.
- The first successful mainframe computer is invented by IBM. Mainframe computer's speed is comparatively less than Supercomputers.

Classification of Computers

Classification of Computers based on size:

Minicomputer:

- A minicomputer is a type of computer that possesses most of the features and capabilities of a large computer but is smaller in physical size.
- A minicomputer fills the space between the mainframe and microcomputer.
- Minicomputers are mainly used as small or mid-range servers operating business and scientific applications. However, the use of the term minicomputer has diminished and has merged with servers.
- A minicomputer may also be called a mid-range computer.
- Examples are IBM AS, Prime series, HP 9000, PDP 11, IBM 8000 series.

Classification of Computers

Classification of Computers:

BASIS FOR COMPARISON	SUPERCOMPUTER	MAINFRAME COMPUTER
Basic	Supercomputers fastly perform large and complex mathematical computations.	Mainframe computers act as a server, stores large database and serve a large number of users simultaneously.
Invention	The first successful Supercomputer was invented by Seymour Cray in the year 1976 Cray 1.	IBM invented the first successful mainframe computer and is still a dominant company for producing the mainframe computers.
Speed	The supercomputer can execute billions of floating point operations in a second.	Mainframe computers can execute millions of instruction simultaneously.
Size	Supercomputers are the largest computers in the world.	Mainframe computers are also large computers but somewhat smaller than supercomputer.
Expense	Supercomputers are the most expensive computers of the worlds.	Mainframe computers are also expensive but less than supercomputers.
Operating system	The modern supercomputers have Linux operating system and derivative variants of Linux operating system.	Mainframe computer has the ability to run multiple operating system. simultaneously.

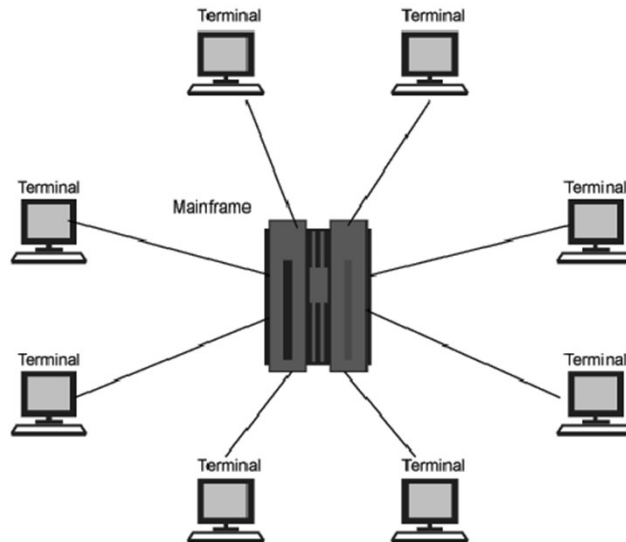
Classification of Computers

Classification of Computers:

Mainframe	Minicomputer
In mainframe computer, large size of disk is used.	While in minicomputer, small size of disk is used.
Mainframe computers have large memory storage.	While minicomputers have small or less memory storage than mainframe computer.
The processing speed of mainframe computer is faster than minicomputer.	While the processing speed of minicomputer is slower than mainframe computer.
Mainframe computer is costlier than minicomputers.	Whereas supercomputer's cost is less or it is Inexpensive.
The first microcomputer was invented by the team leader Bill Pentz .	The first successful mainframe computer is invented by IBM.
Mainframe computers support thousand or millions of users simultaneously.	Whereas minicomputers support hundreds of users at a time.

Classification of Computers

Mainframe Computer:



Classification of Computers

Classification of Computers based on size:

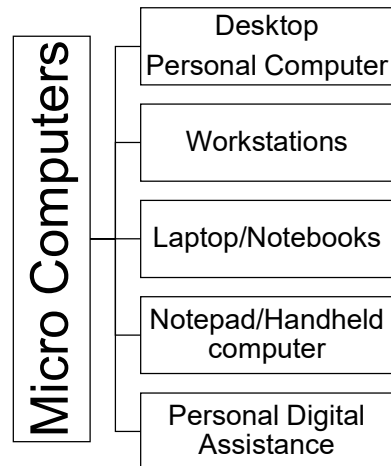
Network Computer:

- A network computer is an inexpensive personal computer designed for a centrally-managed network where data are stored and updated on a network server.
- Network computer doesn't have a disk drive, CD-ROM drive or expansion slots.
- A network computer depends on network servers for processing power and data storage.

Classification of Computers

Classification of Computers based on size:

Micro computer:



Classification of Computers

Classification of Computers based on size:

Micro computer:

1. Desktop/Personal Computer

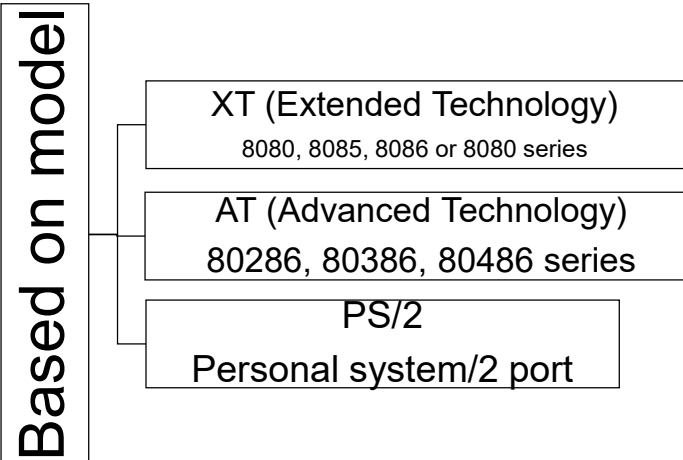
A desktop computer is a personal computer designed for regular use at a single location on or near a desk or table due to its size and power requirements.



Classification of Computers

Classification of Computers based on model:

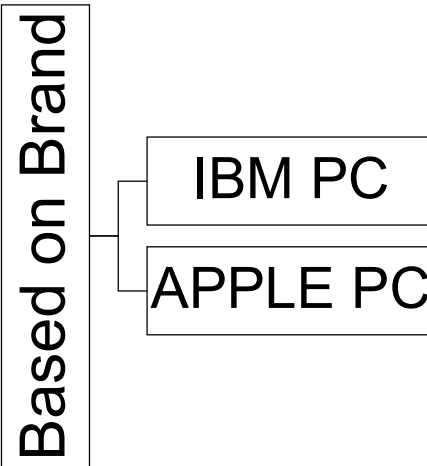
Classification of digital computers on the basis of their model are like:



Classification of Computers

Classification of Computers based on model:

Classification of digital computers on the basis of brand/Manufacturer are like:



Questions

- Q1. Highlight the differences between microcomputer, minicomputer, mainframe computer and supercomputer.

The Computer System

The Computer System:

A system is an organized collection of parts (or subsystems) that are highly integrated to accomplish an overall goal.

A computer system is a set of integrated devices that input, output, process, and store data and information.

COMPUTER SYSTEM is a collection of entities (hardware, software, data and users) that are designed to receive, process, manage and present information in a meaningful format.

The computer system consists of four parts:

1. Computer Hardware
2. Computer Software
3. Data
4. Users: Skinware, liveware, humanware or peopleware.



The Computer System

The Computer System: The computer system consists of four parts

1. **Computer Hardware** - Are physical parts/ intangible parts of a computer. e.g. Input devices, output devices, central processing unit and storage devices
2. **Computer Software** - Software is a set of instructions that tells the computer about the tasks to be performed and how these tasks are to be performed. Program is a set of instructions, written in a language understood by the computer, to perform a specific task. A set of programs and documents are collectively called software. The hardware of the computer system cannot perform any task on its own and needs to be instructed about the task to be performed. Software instructs the computer about the task to be performed. The hardware carries out these tasks. Different software can be loaded on the same hardware to perform different kinds of tasks.

The Computer System

COMPUTER SYSTEM

3. **Data** - Data are isolated values or raw facts, which by themselves have no much significance. Computer by processing changes the data into value or information.
4. **Users** - The user commands the computer system to execute on instructions. Users are people who write computer programs or interact with the computer. They are also known as skinware, liveware, humanware or peopleware. Programmers, data entry operators, system analyst and computer hardware engineers fall into this category.

The Computer System

COMPUTER SYSTEM:

- Hardware and software exist together to make up the actual computer. The humanware component adds in the human factor in order to bring the whole computer into functional and productive existence.
- Each of these components is necessary in order to enable meaningful productivity. If installed separately, each may be useful to an extent but incapable of achieving complete computing potential. Hardware and software need the human factor in order to make input and connectivity possible.

The Computer System

Components of Computer Hardware:

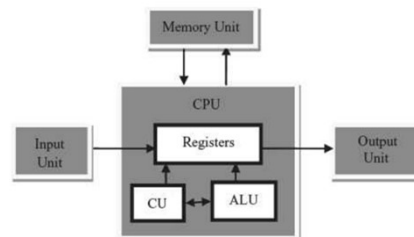
The computer system hardware comprises of three main components:

1. Input/Output (I/O) Unit,
2. Central Processing Unit (CPU), and
3. Memory Unit.

The I/O unit consists of the input unit and the output unit. CPU performs calculations and processing on the input data, to generate the output.

The memory unit is used to store the data, the instructions and the output information.

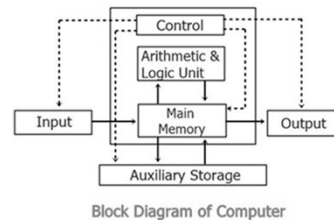
The typical interaction among the different components of the computer.



The Computer System

Components of Computer Hardware:

Input/Output Unit: The user interacts with the computer via the I/O unit. The Input unit accepts data from the user and the Output unit provides the processed data i.e. the information to the user. The Input unit converts the data that it accepts from the user, into a form that is understandable by the computer. Similarly, the Output unit provides the output in a form that is understandable by the user. The input is provided to the computer using input devices like keyboard, trackball and mouse. Some of the commonly used output devices are monitor and printer.



The Computer System

Components of Computer Hardware:

Central Processing Unit CPU controls, coordinates and supervises the operations of the computer. It is responsible for processing of the input data. CPU consists of Arithmetic Logic Unit (ALU) and Control Unit (CU).

- ALU performs all the arithmetic and logic operations on the input data.
- CU controls the overall operations of the computer i.e. it checks the sequence of execution of instructions, and, controls and coordinates the overall functioning of the units of computer.

Additionally, CPU also has a set of registers for temporary storage of data, instructions, addresses and intermediate results of calculation.

The Computer System

Components of Computer Hardware:

Memory Unit: Memory unit stores the data, instructions, intermediate results and output, temporarily, during the processing of data. This memory is also called the main memory or primary memory of the computer. The input data that is to be processed is brought into the main memory before processing. The instructions required for processing of data and any intermediate results are also stored in the main memory. The output is stored in memory before being transferred to the output device. CPU can work with the information stored in the main memory. Another kind of storage unit is also referred to as the secondary memory of the computer. The data, the programs and the output are stored permanently in the storage unit of the computer. Magnetic disks, optical disks and magnetic tapes are examples of secondary memory.

Q2. List the steps in the working of the computer.

Q3. Define Computer System. Explain the parts of computer system.

Applications of computers

Applications of computers:

Today computers find widespread applications in all activities of the modern world. Some of the major application areas include:

- Scientific, Engineering and Research
- Business
- Medicine
- Information
- Education
- Government
- Games and Entertainment

Applications of computers

Applications of computers:

- **Uses of Computers in Education**
 - Benefits of CBT
 - Computer Aided Learning (CAL)
 - Distance Learning
 - Online Examination
 - Online Training Resources
- **Uses of computers in Medical Field**
 - Hospital Management System
 - Patient History
 - Patients Monitoring
 - Life Support Systems
 - Diagnosis Purpose

Applications of computers

Applications of computers:

- Computers in Business
 - Marketing
 - Stock Exchange
- Government
- Use of computer at Home
 - Computer Games
 - Working from Home
 - Entertainment
 - Information

Applications of computers

Applications of computers:

- The list of applications of computers is so long that it is not possible to discuss all of them here.
- In addition to the applications of the computers discussed above, computers have also proliferated into areas like banks, investments, stock trading, accounting, ticket reservation, military operations, meteorological predictions, social networking, business organizations, police department, video conferencing, telepresence, book publishing, web newspapers, and information sharing.